Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The wavefronts of a light wave travelling in vacuum are given by $x + y + z = c$. The angle made by the direction of propagation of light with the X-axis is

(A) 0°

(B) 45°

(C) 90°

(D) $\cos^{-1}(1/\sqrt{3})$

Q.2

Huygen's concept of secondary wave

(A) allows us to find the focal length of a thick lens

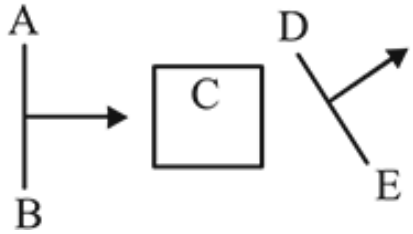
(B) is a geometrical method to find a wavefront

(C) is used to determine the velocity of light

(D) is used to explain polarisation

Q.3

A wavefront AB passing through a system C emerges as DE. The system C could be



(A) a slit

(B) a biprism

(C) a prism

(D) a glass slab

Q.4

The phase difference between incident wave and reflected wave is 180° when light ray

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(A) enters into glass from air

(B) enters into air from glass

(C) enters into glass from diamond

(D) enters into water from glass

Q.5

A beam of monochromatic light is refracted from vacuum into a medium of refractive index 1.5. The wavelength of refracted light will be

(A) dependent on intensity of refracted light

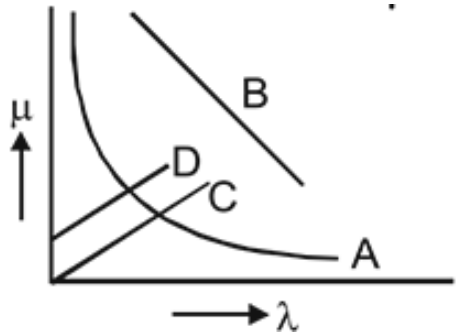
(B) same

(C) smaller

(D) larger

Q.6

The correct curve between refractive index μ and wavelength λ will be



(A) A

(B) D

(C) B

(D) C

Q.7

From a medium of index of refraction n_1, monochromatic light of wavelength λ is incident normally on a thin film of uniform thickness L (where $\hat{L} > 0.1\lambda$) and index of refraction n_2. The light transmitted by the film travels into a medium with refractive index n_3. The value of minimum film thickness when maximum light is transmitted if $(n_1 < n_2 < n_3)$ is

(A) $\frac{n_1\lambda}{2n_2}$

(B) $\frac{n_1\lambda}{4n_2}$

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(C) $\frac{\lambda}{4n_2}$

(D) $\frac{\lambda}{2n_2}$

Q.8

A glass plate 0.40 micron thick is illuminated by a beam of white light normal to the plate. The refractive index of glass is 1.50 and the limits of the visible spectrum are $\lambda_{\mathrm{v}} = 4000\mathring{A}$ and $\lambda_{\mathrm{R}} = 7000\mathring{A}$. The wavelengths that get intensified in the reflected beam are

(A) $4800\mathring{A}$ and $6700\mathring{A}$

(B) $4800\mathring{A}$ and $5200\mathring{A}$

(C) $4800\mathring{A}$ only

(D) $5200\mathring{A}$ only

Q.9

A certain region of a soap bubble reflects red light of wavelength $\lambda = 650\ \mathrm{nm}$. What is the minimum thickness that this region of the soap bubble could have? Take the index of reflection of the soap film to be 1.41.

(A) $1.2 \times 10^{-7}\ \mathrm{m}$

(B) $650 \times 10^{-9}\ \mathrm{m}$

(C) $120 \times 10^{7}\ \mathrm{m}$

(D) $650 \times 10^{5}\ \mathrm{m}$

Q.10

Two identical light waves, propagating in the same direction, have a phase difference δ. After they superpose, the intensity of the resulting wave will be proportional to

(A) $\cos\delta$

(B) $\cos(\delta/2)$

(C) $\cos^2(\delta/2)$

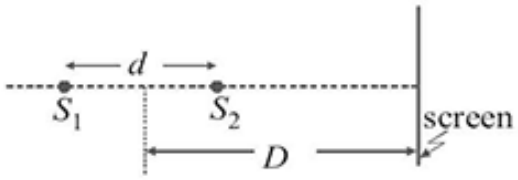
(D) $\cos^2\delta$

Q.11

Two coherent point sources S_1 and S_2 are separated by a small distance d as shown. The fringes

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obtained on the vertical screen will be :



(A) points

(B) straight bands

(C) concentric circles

(D) semicircles

Q.12

Which of the following is not an essential condition for interference

(A) The two interfering waves must be propagated in almost the same direction or the two interfering waves must intersect at very small angle,

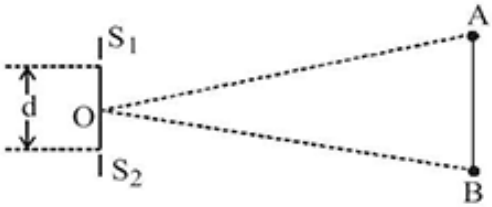
(B) The waves must have the same period and wavelength,

(C) The amplitude of the two waves must be equal,

(D) The two interfering beams of light must originate from the same source.

Q.13

Figure shows two coherent sources S_1 and S_2 vibrating in same phase. AB is an irregular wire lying at a far distance
from the sources S_1 and S_2. Let $\frac{\lambda}{d} = 10^{-3}$ and $\angle BOA = 0.12^\circ$.
How many bright spots will be seen on the wire, including points A and B ?



(A) 4

(B) 3

(C) 1

(D) 5

Q.14

A double slit of separation 0.1 cm is illuminated by white light. A coloured interference

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pattem is formed on a screen $100\ \text{cm}$ away. If a pin hole is located in this screen at a distance of $2\ \text{mm}$ from the central fringe, the wavelengths in the visible spectrum which will be absent in the light transmitted through the pin-hole are

(A) $5714\mathring{A}$ and $4444\mathring{A}$

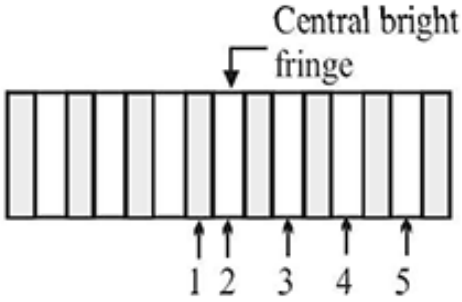
(B) $6000\mathring{A}$ and $5000\ \text{A}$

(C) $5500\mathring{A}$ and $4500\mathring{A}$

(D) $5200\mathring{A}$ and $4200\mathring{A}$

Q.15

The figure shows the interference pattern obtained in a double-slit experiment using light of wavelength $600\ \text{nm}$. $1, 2, 3, 4$ and 5 are marked on five fringes.



The third order bright fringe is

(A) 1

(B) 3

(C) 4

(D) 5

Q.16

A possible means for making an airplane invisible to radar is to coat the plane with an anti reflective polymer. Radar waves have a wavelength of $3.00\ \text{cm}$ and the index of refraction of the polymer is $n = 1.50$. How thick would you make the coating?

(A) $1.50\ \text{cm}$

(B) $3.00\ \text{cm}$

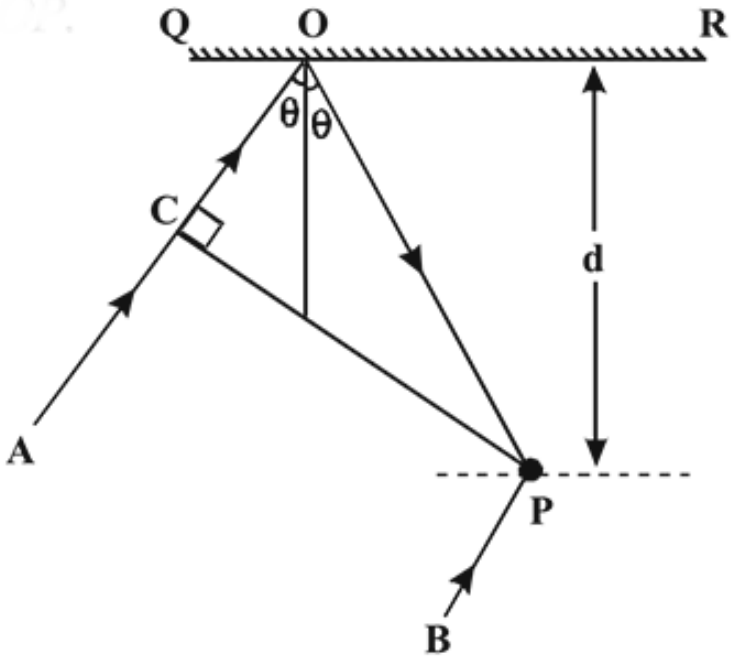
(C) $0.50\ \text{cm}$

(D) None of these

Q.17

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In the adjacent diagram, CP represents a wavefront and $AO\&BP$, the corresponding two rays. Find the condition on θ for constructive interference at P between the ray BP and reflected ray



(A) $\cos\theta = 3\lambda/2d$

(B) $\cos\theta = \lambda/4d$

(C) $\sec\theta - \cos\theta = \lambda/d$

(D) $\sec\theta - \cos\theta = 4\lambda/d$

Q.18

Two slits separated by a distance of $1\ \mathrm{mm}$ are illuminated with red light of wavelength $6.5 \times 10^{-7}\ \mathrm{m}$. The interference fringes are observed on a screen placed $1\ \mathrm{m}$ from the silts. The distance of the third dark fringe from the central fringe will be equal to :

(A) $0.65\ \mathrm{mm}$

(B) $1.30\ \mathrm{mm}$

(C) $1.62\ \mathrm{mm}$

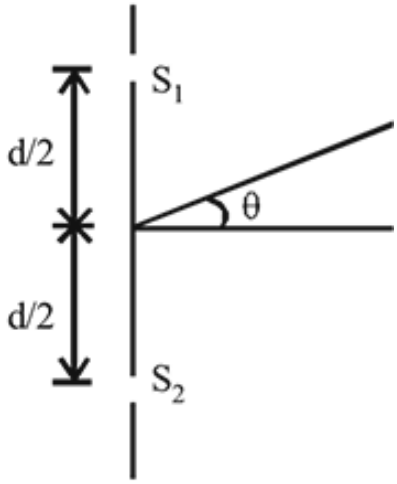
(D) $1.95\ \mathrm{mm}$

Q.19

In an interference arrangement similar to Young's double-slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency $10^6\ \mathrm{Hz}$. The sources are synchronized to have zero phase difference. The slits are separated by a distance $\mathrm{d} = 150.0\ \mathrm{m}$. The intensity $I(\theta)$ is measured as a function of θ, where θ is defined as shown. If I_0 is the maximum intensity, then $I(\theta)$ for

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$0 \leq \theta \leq 90^{\circ}$ is given by



(A) $I(\theta) = I_o/2$ for $heta a$=30

(B) $I(\theta) = I_o/4$ for $heta a$=90

(C) $I(\theta) = I_o$ for $heta a$=0

(D) $I(\theta)$ is constant for all values of θ.

Q.20

If white light is used in the Newton's rings experiment, the colour observed in the reflected light is complementary to that observed in the transmitted light through the same point. This is due to

(A) 90° change of phase in one of the reflected waves

(B) 180° change of phase in one of the reflected waves

(C) 145° change of phase in one of the reflected waves

(D) 45° change of phase in one of the reflected waves

Answers & Solutions

Q.1 Answer:

$\cos^{-1}(1/\sqrt{3})$

Solution:

Let any $\overrightarrow{\mathrm{R}}$ **its components are**

$\overrightarrow{\mathrm{R}} = \overrightarrow{\mathrm{R}}\mathrm{x} + \overrightarrow{\mathrm{R}}\mathrm{y} + \overrightarrow{\mathrm{R}z}$

with $|\overrightarrow{\mathrm{R}}| = \overrightarrow{\mathrm{R}} = |\overrightarrow{\mathrm{R}}| = \overrightarrow{\mathrm{R}} = \sqrt{\mathrm{Rx}^2 + \mathrm{Ry}^2 + \mathrm{Rz}}$

$\& \cos\theta_{\mathrm{x}} = \frac{\mathrm{Rx}}{\mathrm{R}}, \cos\theta_{\mathrm{y}} \frac{\mathrm{Ry}}{\mathrm{R}}, \cos\theta_{\mathrm{z}} = \frac{\mathrm{Rz}}{\mathrm{R}}$

there $\cos\theta_x, \cos\theta_y$ **and** $\cos\theta_z$ **one called direction cosines.**

Hence $x + y + z = c \quad (= \vec{R})$

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So, magnitude of $c = \sqrt{I^2 + I^2 + I^2} = \sqrt{3}$
and $\cos\theta_x = \frac{1}{\sqrt{3}}$

Q.2 Answer:

is a geometrical method to find a wavefront

Solution:

Huygen's principle gives us a geometrical method of tracing a wavefront.

Q.3 Answer:

a prism

Solution:

A slit would give divergent; a biprism would give double; a glass slab would give a parallel wavefront. Edge is downward.

Q.4 Answer:

enters into glass from air

Solution:

When light reflects from denser surface phase change of π **occurs.**

Q.5 Answer:

smaller

Solution:

From $\mu = \frac{c}{v} = \frac{n\lambda_v}{n\lambda_m}, \lambda_m = \frac{\lambda_v}{\mu}$
Here, c **= velocity of light in vacuum and** v **= velocity of light in medium** m **= refractive index of the medium. Hence, wavelength in medium** $(\lambda_m) < \lambda_v (\because$ $\mu > 1$**, given** $)$ **So, the required wavelength decreases.**

Q.6 Answer:

A

Solution:

According to Cauchy's formula $\mu = A + \frac{B}{\lambda^2} + \ldots$ **Therefore, as** λ **increases, u decreases. Curve A is correct.**

Q.7 Answer:

$\frac{n_1\lambda}{4n_2}$

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Solution:

Equation of path difference form maxima in transmission (or weak reflection) $\Delta P_{opt} = 2n_2L = \frac{\lambda_{\text{vacuum}}}{2}, \frac{3\lambda_{\text{vacuum}}}{2} \ldots\ldots$

$\Rightarrow 2\left(\frac{n_2}{n_1}\right)L = \frac{\lambda}{2}, \frac{3\lambda}{2}, \ldots.. \Rightarrow L = \frac{\lambda}{4n_2}$

(notice that $\lambda =$ **wavelength in medium is related to** λ_{vacuum} **as,** $\lambda_{\text{vacuum}} = n_1\lambda$**)**

Q.8 Answer:

$4800\mathring{A}$ only

Solution:

In Reflected light; $2\mu t = (2n+1)\frac{\lambda}{2}$

$$\lambda = \frac{4\mu t}{2n+1} \Rightarrow \frac{4\times 400\times 10^{-9}\times 1.5}{2n+1}$$
$$\lambda = \frac{2400nm}{2n+1}$$
$$\text{for } n=1 \quad \lambda_1 = 800 \text{ nm} \Rightarrow 8000^{\circ}$$

$$\text{for } n=2 \quad \lambda_2 = 480 \text{ nm} \Rightarrow 4800 \text{ A}$$
$$\text{for } n=3 \quad \lambda_3 = 342.85 \text{ nm} \Rightarrow 3428.5 \text{ A}$$

Hence as only $\lambda_2 = 4800$ **A lies in the spectrum ranging 4000 A-700OA Only 4800 A Wavelength will get intensified.**

Q.9 Answer:

1.2×10^{-7} m

Solution:

There is air on both sides of the soap film. $\therefore$ **the reflections of the light produce a net** 180° **phase shift. The condition for bright fringes is** $2t = (m+1/2)\lambda_{\text{film}}$

$t = \frac{(m+1/2)\lambda_{\text{film}}}{2} = \frac{(m+1/2)\lambda}{2n}$

$= \frac{(1/2)\left(650\times 10^{-9}\text{ m}\right)}{2(1.41)} = 1.2 \times 10^{-7}\text{ m}$

Q.10 Answer:

$\cos^2(\delta/2)$

Solution:

Here $A^2 = a_1^2 + a_2^2 + 2a_1a_2\cos\delta$

$\because a_1 = a_2 = a$

$\therefore A^2 = 2a^2(1+\cos\delta) = 2a^2\left(1+2\cos^2\frac{\delta}{2}-1\right)$

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$\Rightarrow \mathrm{A}^2 \propto \cos^2 \frac{\delta}{2}$
Now, $I \propto A^2 \quad \therefore I \propto A^2 \propto \cos^2 \frac{\delta}{2}$
$\therefore \mathrm{I} \propto \cos^2 \frac{\delta}{2}$

Q.11 Answer:

concentric circles

Solution:

Correct option is (c)concentric circles

for a given path difference many points can be found on the screen and locus of all these points are semicircles. hence, fringes will be in concentric circle forms.

Q.12 Answer:

The amplitude of the two waves must be equal,

Solution:

The amplitude of the waves can be anything for interference.

Q.13 Answer:

3

Solution:

Angular width $= \frac{\lambda}{\mathrm{d}} = 10^{-3}$ **(given)**
∴ No. of fringes within 0.12° **will be**
$\mathrm{n} = \frac{0.12 \times 2\pi}{360 \times 10^{-3}} \cong [2.09]$
∴ The number of bright spots will be three.

Q.14 Answer:

$5714\mathring{A}$ and $4444\mathring{A}$

Solution:

Absent wavelengths correspond to interference minima
$\therefore d \sin^{\theta} = (2n-1)\frac{\lambda}{2}$
$\Rightarrow d.\frac{\mathrm{y}}{\mathrm{D}} = (2n-1)\frac{\lambda}{2}$
$\Rightarrow \lambda = \frac{2\phi y}{D(2n-1)} = \frac{40{,}000}{2n-1} \& = 13333, 8000, 5714, 4444, 3636, \ldots.$

Q.15 Answer:

5

Solution:

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(d)Order of the fringe can be counted on either side of the central maximum. For example, Fringe no.3 is first-order brigth fringe.

Q.16 Answer:

0.50 cm

Solution:

Treating the anti-reflectance coating like a camera-lens coating, one can obtain for its thickness t :

$$2t = \left(m + \frac{1}{2}\right)\frac{\lambda}{n}, m = 0, \pm 1, \pm 2, \ldots\ldots\ldots$$

Condition (1) is the condition for the destructive interference for the normally incident and reflected electromagnetic wave. The minimum thickness of the coat refers to $m = 0$. This gives for the thickness:

$t = \frac{\lambda}{4n} = \frac{3.00}{4(1.5)}\text{cm} = 0.500$ cm

This anti-reflectance coating could be easily countered by changing the wavelength of the radar to 1.50 cm now creating maximum reflection.

Q.17 Answer:

$\cos\theta = \lambda/4d$

Solution:

Correct option is (b) $\cos\theta = \lambda/4\ \mathrm{d}$

From geometry,

$\mathrm{OP} = \frac{\mathrm{d}}{\cos\theta}$

Path difference between the rays $= \mathrm{CO} + \mathrm{OP} = \frac{\mathrm{d}}{\cos\theta}\cos 2\theta + \frac{\mathrm{d}}{\cos\theta} = 2\ \mathrm{d}\cos\theta$

Since there is a phase change by π due to reflection from optically denser medium, condition for constructive interference is

Path difference $= \frac{\lambda}{2}$

$\Longrightarrow \cos\theta = \frac{\lambda}{4\ \mathrm{d}}$

Q.18 Answer:

1.62 mm

Solution:

Given $d = 1$ mm, $\lambda = 6.5 \times 10^{-7}$ m, $D = 1$ m.

For n^{th} dark fringe $y_n = \left(\frac{2n-1}{2}\right)\frac{D\lambda}{d}$ **For third fringe,** $y_3 = \left(\frac{2\times 3-1}{2}\right) \times \frac{1\times 6.5\times 10^{-7}}{1\times 10^{-3}} =$

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1.625 mm.

Q.19 Answer:

$I(\theta) = I_o$ for $heta$=0

Solution:

We know that $I(\theta) = I_0 \cos^2 \frac{\delta}{2}$ **where** $\delta = \frac{2\pi d \tan\theta}{\lambda}$

$I(\theta) = I_0 \cos^2 \left(\frac{\pi d \tan\theta}{\lambda}\right)^2 = I_0 \cos^2 \left(\frac{\pi \times 150 \times \tan\theta}{3\times 10^8/10^6}\right)$

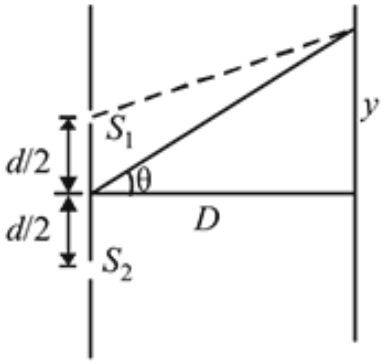
$= I_0 \cos^2 \left(\frac{\pi}{2} \tan\theta\right)$

For $\theta = 30^\circ$;

$I(\theta) = I_o \cos^2 \left(\frac{\pi}{2\sqrt{3}}\right)$

For $\theta = 90^\circ; I(\theta) = I_o \cos^2(\infty)$

For $\theta = 0^\circ I(\theta) = I_0$



$I(\theta)$ **is not constant. Alternatively, when** θ **is zero the path difference between wave originating from** S_1 **and that from** S_2 **will be zero. This corresponds to a maxima.**

Q.20 Answer:

180° change of phase in one of the reflected waves

Solution:

The rings observed in reflected light are exactly complementary to those seen in transmitted light. Corresponding to every dark ring in reflected light there is a bright ring in transmitted light. The ray reflected at the upper surface of the air-film suffers no phase change while the ray reflected internally at the lower surface suffers a phase change of π

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